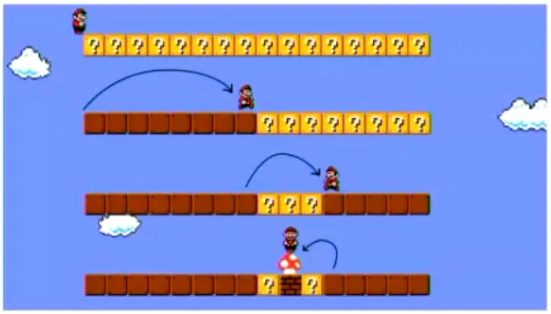
Mergesort Recurrence Relation

$T(n) = 2T(n/2) + n$ $\quad T(n/2) = 2T(n/4) + n/2$

$= 2(2T(n/2^2) + n/2) + n$

$= 2^2 T(n/2^2) + 2n$



Consider the recursive version of binary search.

- ~~What would the pseudocode look like?~~
- What is the **recurrence relation**?
  - Write it down by a brief explanation on what each term explicitly means in plain English.
- What **is the runtime**?
  - Step by step, calculate the runtime. Show your work!

$T(n) = T(n/2) + 4$ $\quad T(n/2) = T(n/4) + 4$

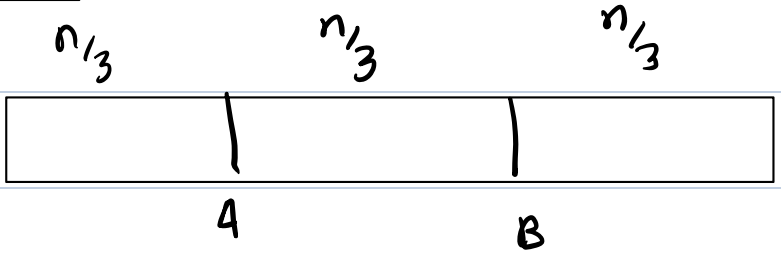
$L_2 = T(n/4) + 4 + 4$ $\quad T(n/4) = T(n/8) + 4$

$L_3 = T(n/8) + 3 \cdot 4$

$L_k = T(n/2^k) + 4k$

$T(n/2\log_2 n) + 4\log_2 n$

$= \boxed{\theta \log_2 n}$



Compare val to A

if greater check B

Psudo code

```
Tri search( Array, start, end, val)
{ if( start == end)
```

return false

$A = \frac{end - start}{3}$

$B = \frac{end - start}{3} \cdot 2$

if (Array [a] == val) return True

else if (Array [a] > val) Tri search (Array [n-1] val)

if (Array [b] == val) return True

else if (Array [b] > val) Tri search (Array [a+1, b-1] val

else Trisearch (Array b+1, end val)

}

Reccurance Relation

$T(n) = T(\frac{n}{3}) + 7 \qquad T(\frac{n}{3}) = T(\frac{n}{9}) + 7$

$= T(\frac{n}{3^2}) + 7 \cdot 2$

$L_k = T(\frac{n}{3^k}) + 7k \qquad \frac{n}{3^k} = 1$

$= 1 + 7 \log_3 n \qquad \log_3 n = k$

$= \Theta \log_3 n$

Tri search is slower than binary search because the constants are less